

COMP1814 - Statistical Techniques with R

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Dataset: birthwt_student_001313271.csv

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Section 1 - Exploring and Understanding the Data

The dataset contains 100 observations and 10 variables generated from the MASS birthwt dataset using `set.seed(001313271)`. Three randomisation steps were applied: (A) a random 100-row sample, (B) Gaussian noise ($SD = 20g$) added to birth weights, and (C) smoking status shuffled across all observations. Step C is critical since it deliberately destroys any real-world association between smoking and birth weight, shaping all subsequent results throughout this report.

Variable	Data Type	Description
birthwt.below.2500	Factor (2 levels: "no"/"yes")	Low birth weight indicator (<2500g)
mother.age	int	Mother's age at pregnancy
factorA	int	Randomised label (could be mother weight)
factorB	Factor (3 levels: "black", "white", "other")	Randomised label (could be race)
factorC	Factor (2 levels: "no"/"yes")	Randomised label identified as smoking status
previous.prem.labor	int	Number of previous premature labours
hypertension	Factor (2 levels: "no"/"yes")	Maternal hypertension during pregnancy
uterine.irr	Factor (2 levels: "no"/"yes")	Uterine irritability (links to preterm labour)
physician.visits	int	First-trimester physician visit count
birthwt.grams	num	Birth weight of the baby in grams (main outcome variable)

Identifying factorC: `str(birthwt)` confirmed factorC was the only remaining yes/no factor unmapped to a known variable, confirming it as mother.smokes. Converting to `factor()` was essential, without it, `t.test(birthwt.grams ~ factorC)` would not execute.

```
#Converting categorical variables into factors
library(dplyr)
birthwt <- birthwt %>%
  mutate(
    birthwt.below.2500 = factor(birthwt.below.2500),
    factorB = factor(factorB),
    factorC = factor(factorC),
    hypertension = factor(hypertension),
    uterine.irr = factor(uterine.irr))

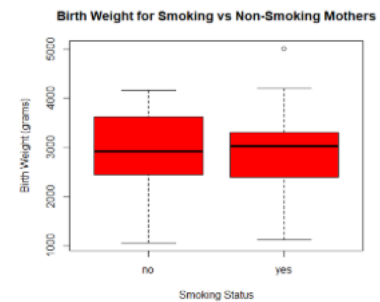
#Mean, median and SD for numerical variables
birthwt %>%
  summarise(across(where(is.numeric), list(
    mean = mean,
    median = median,
    sd = sd)))
```

Variable	Mean	Median	SD	Observation
mother.age	23.87	23	5.647302	Most mothers in their early 20s
factorA	131.32	123.5	30.67947	High SD reflects variability in maternal weight (lbs)
previous.prem.labor	0.19	0	0.5064234	Majority had no previous premature labours
physician.visits	0.89	1	1.153782	Below 1 visit on average, low antenatal engagement in this sample
birthwt.grams	2935.525	2924.3	753.9738	Mean below 3000g, high SD indicates wide variability, Gaussian noise adds little

The average age of the mothers in my dataset rounds up to 24 years, which is lower than the current average of 31 years in the UK (ONS, 2023). The average birth weight of the babies is under 3kg, which is lower than the global average of 3.2 - 3.4kg (WHO, 2023). Most of the mothers in my dataset did not experience previous premature labours, which mirrors global trends. The number of early pregnancy visits to the doctor is lower than the current recommendation of the NHS (2023), this shows that my dataset is not representative of current practice. All the statistics show that my dataset should not be taken as representative of current populations.

Section 2 – Comparing Means Between 2 Groups

```
#SECTION 2
#Boxplot between smoking and non-smoking mothers
boxplot(birthwt.grams ~ factorC, data = birthwt,
        xlab = "Smoking Status",
        ylab = "Birth weight (grams)",
        main = "Birth weight for Smoking vs Non-Smoking Mothers",
        col = "red")
#Mean, SD and standard error of birth weight for smoking vs. non-smoking mothers
birthwt %>%
  group_by(factorC) %>%
  summarise(
    mean = mean(birthwt.grams),
    sd = sd(birthwt.grams),
    se = sd / sqrt(n()))
#Two-sample t-test to assess whether smoking significantly affects birth weight
t.test(birthwt.grams ~ factorC, data = birthwt)
```



The boxplot shows that babies born to non-smoking mothers tend to be heavier than those born to smoking mothers', the interquartile box for non-smokers sits higher up the chart and the median is also higher. The smokers' group has a wider box meaning birth weights are more spread out and less consistent within that group. There is also one unusually high birth weight baby in the smokers' group, given that there are fewer smokers than non-smokers in the dataset, a single outlier has more influence in a smaller group. The smoking labels in this dataset were randomly reassigned during data generation, the "smoker" and "non-smoker" groups were not formed based on actual smoking behaviour, the visual difference between the two groups is most likely the result of chance rather than a real effect of smoking.

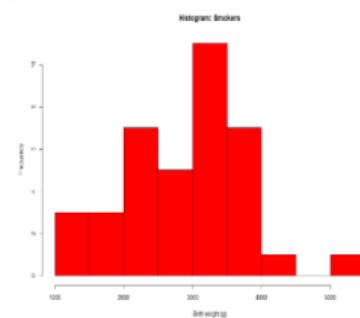
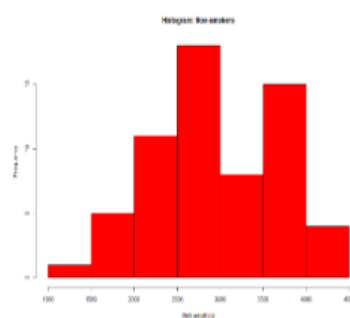
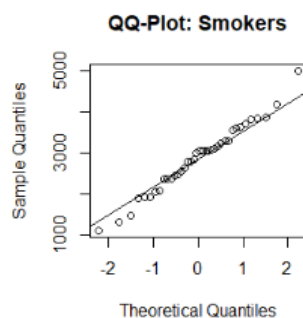
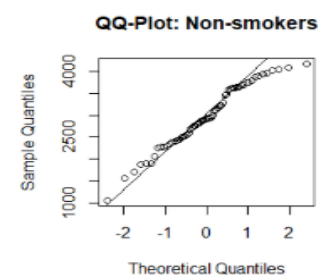
factorC	mean	sd	se
no	2970.	714.	90.7
yes	2880.	821.	133.

The two groups have noticeably different spreads (SD = 714g vs 821g). R's t-test automatically accounts for this imbalance in its calculation, making a separate F-test or Levene's test unnecessary here. The higher variability in the smokers' group and the larger SE reflect the smaller group size (n=38 vs n=62). However, any interpretation of these trends as being representative of actual smoker behaviour is invalid since the smoker/non-smoker classification was randomly assigned.

Statistic	Value	Interpretation
t-statistic	0.56054	The difference between the group averages is very small compared with how much the birth weights vary overall.
p-value	0.5769	Higher than 0.05, meaning the difference is not statistically significant.
95% Confidence Interval	-231.1192g to 411.8248g	The confidence interval shows the true difference could be higher, lower, or zero. We cannot be sure there is any real effect.
Significant at 5%?	No	There is not enough evidence to conclude smoking affects birth weight in this dataset as the p-value is > 0.05.

Section 3 – Checking Normality and Alternative Tests

```
#SECTION 3
#QQ plot
bw_no <- birthwt$birthwt.grams[birthwt$factorC == "no"]
bw_yes <- birthwt$birthwt.grams[birthwt$factorC == "yes"]
qqnorm(bw_no, main = "QQ-Plot: Non-smokers")
qqline(bw_no)
qqnorm(bw_yes, main = "QQ-Plot: Smokers")
qqline(bw_yes)
#Histogram
hist(bw_no, main = "Histogram: Non-smokers", xlab = "Birth weight (g)", col = "red")
hist(bw_yes, main = "Histogram: Smokers", xlab = "Birth weight (g)", col = "red")
#wilcoxon rank-sum test
wilcox.test(bw_no, bw_yes)
```



Normality Assessment Group	QQ-plot	Histogram	Verdict
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Non-smokers	Moderate tail bending	Roughly symmetric, slight right skew	Borderline
Smokers	Strong upper tail deviation	Right-skewed, longer lower tail	Violated

Test Comparison Test	Statistic	p-value	Conclusion
Two-sample t-test	$t = 0.5605$	0.5769	No significant difference
Wilcoxon rank-sum	$W = 1252$	0.6017	No significant difference

Normality is violated due to non-linear QQ plots and skewed histogram of both groups. Hence, a Wilcoxon rank-sum test is run because it doesn't require normality and is more robust to skewness and non-linear tails since it compares the order of values instead.

The p value of the Wilcoxon and t-test is both above 0.05 hence, both agree that there's no statistically significant difference in birth weight between two groups. My conclusion of no significant difference is the same regardless of which test is used, there is no evidence of a real difference in birth weight of babies between smokers and non-smoking mothers in my dataset. This result is also expected because, in this case, the smoking variable was randomly shuffled during data generation, therefore, these two groups are random partitions of the same distribution and are statistically equivalent regardless of which test is used.

Section 4 – Hypothesis Testing for Categorical Data

```
#SECTION 4
#2x2 contingency table
table_LBW_smoke <- table(birthwt$birthwt.below.2500, birthwt$factorc)
table_LBW_smoke
#Fisher's test
fisher.test(table_LBW_smoke)
#Chi-squared test
chisq.test(table_LBW_smoke)
```

2×2 Contingency Table Low birth weight?	Non-smoker (n=62)	Smoker (n=38)	Row total
No	43 (69.4%)	25 (65.8%)	68
Yes	19 (30.6%)	13 (34.2%)	32
Total	62 (100%)	38 (100%)	100

Test	p-value	Extra output	Conclusion
Fisher's exact	0.8257	OR = 1.17 95% CI: 0.4506959 - 3.0150218	No significant association
Chi-squared	0.8806	$X^2 \approx 0.022$	Confirms Fisher's result

Low birth weight rates are nearly identical (difference of less than 4 percentage points). Both tests produce p-value > 0.05 and the Fisher's odds ratio (1.17) with a CI spanning 0.45–3.02 is large hence statistically meaningless. This result reflects the shuffled smoking assignment. My conclusion still doesn't change.

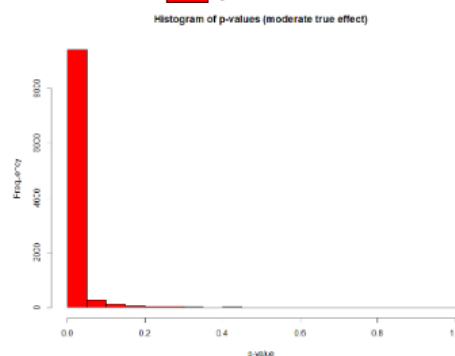
Section 5 – Simulation and Understanding P-Values

```

set.seed(12345)
#SECTION 5
n <- 100 #100 samples for simulation
#Control and treatment group (with the same mean so no effect)
control_group <- rnorm(n, mean = 0, sd = 1)
treat_group_no_effect <- rnorm(n, mean = 0, sd = 1)
#T-test of treatment group with no effect
t_test_treat_group_no_effect <- t.test(treat_group_no_effect, control_group)
t_test_treat_group_no_effect$p.value

#Control group
control_group <- rnorm(n, mean = 0, sd = 1)
#Treatment group WITH a real effect
treat_group_effect <- rnorm(n, mean = true_effect, sd = 1)
#Storing p-value from t-test
p_values[i] <- t.test(treat_group_effect, control_group)$p.value}
#Histogram of p-values
hist(p_values,
     breaks = 30,
     main = "Histogram of p-values (moderate true effect)",
     xlab = "p-value",
     col = "red")

```



For the first simulation, the treatment group was set with no real effect and the t-test gave a p-value of 0.1855222. I used `set.seed(12345)` at the start of the simulation to ensure that every random number generated by `rnorm()` was reproducible. If I had not used `set.seed(12345)`, the p-value of the first simulation would have changed every time the code was run because `rnorm()` generates a new set of random values each time it is called. This randomness is why false positives happen; a test will have a low p-value even though there is no real effect in the test because of the randomness of the test results.

When I performed the experiment 10,000 times with a real effect (mean difference = 0.5), instead of the p-values being distributed across the entire 0 to 1 interval, they were tightly clustered around 0, this is because the true effect is being detected by the t-test, this demonstrates statistical power. When the true effect is present, the t-test is much more likely to return a small p-value because the data is no longer random.

Bonus Question

```

#Bonus task
#30 observations sampled
sample_index <- sample(1:nrow(birthwt), 30)
birthwt_small <- birthwt[sample_index, ]
bw_small_no <- birthwt_small$birthwt.grams[birthwt_small$factorC == "no"] #non-smokers
bw_small_yes <- birthwt_small$birthwt.grams[birthwt_small$factorC == "yes"] #smokers
#T-test
t_test_small <- t.test(bw_small_no, bw_small_yes)
t_test_small
#Wilcoxon test
wilcox_small <- wilcox.test(bw_small_no, bw_small_yes)
wilcox_small

```

```

Welch Two Sample t-test

data: bw_small_no and bw_small_yes
t = 0.99408, df = 21.503, p-value = 0.3312
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -290.2898  823.4136
sample estimates:
mean of x mean of y
 3156.227  2889.665

```

Wilcoxon rank sum exact test

```

data: bw_small_no and bw_small_yes
W = 140, p-value = 0.2287
alternative hypothesis: true location shift is not equal to 0

```

Both tests produce wide confidence intervals, less stable p-values and results heavily dependent on which 30 individuals are selected (leading to a loss of power and difficulty in detecting differences, even if they exist). The t-test's normality and equal-variance assumptions are harder to justify with limited data. The Wilcoxon test makes fewer assumptions and is less distorted by extreme values, making it the safer choice when n is small. Overall, the behaviour of both tests highlights how small datasets can produce noisy, uncertain results and why more robust, non-parametric methods are often preferred when n is limited.

REFERENCES

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